



Divya Mehul Rajparia

B.Tech in Engineering Science and Specialisation in Computer Science & Engineering

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Degree	University/Institute	Year	CGPA/ Marks(%)
B.Tech in Engineering Science and Specialisation in Computer Science & Engineering	IIT Hyderabad	2026	9.39
XII (HSC)	Pace Junior Science College	2022	91.50%
X (ICSE)	Lokhandwala Foundation School	2020	98.17%

EXPERIENCE

Machine Learning Research Intern - Lab for Machine Learning and Biomedicine May 2025 - Present
Viterbi School of Engineering, University of Southern California (Advisors: Dr. Ruishan Liu and Dr. Sai Praneeth Karmireddy)

- Automated tumor segmentation in breast ultrasound (BUSI, BUS-UCLM), reducing reliance on manual annotation and improving diagnostic efficiency.
- Adapted and trained a **PyTorch** implementation of MFMSNet (multi-frequency, multi-scale) for breast ultrasound segmentation.
- Designed and implemented a **CycleGAN-based style-transfer** pipeline to reduce domain shift across BUSI and BUS-UCLM datasets and enrich them with more semantically sound samples.
- The richer datasets led to improvements of **DSC by +11.21%** and **IoU by +15.81%** on image segmentation on the BUSI dataset.
- Built a modular data pipeline; engineered training, checkpointing, and metric logging for reproducible experiments.
- Currently developing a privacy-preserving federated feature-alignment system using a UNet encoder and channel-wise mean/variance exchange with differential privacy configuration.

PROJECTS

DES: A Smarter Way to Grow Machine Learning Models Over Time | [Link to Manuscript](#) Jan 2025 - Present
Machine Learning and Vision Lab, IIT Hyderabad (Advisor: Dr. Vineeth N Balasubramanian)

- Proposed **AFCIL** (Assumption-Free Class-Incremental Learning), a setup that performs class-incremental learning without **stored data** and handles **imbalanced tasks** using **pre-trained features**.
- Designed **Dynamic Expert Saturation (DES)**: a two-stage expert system that first specializes on head classes then expands to tail classes to reduce catastrophic forgetting.
- Implemented the research codebase in **PyTorch**, integrating a frozen **CLIP backbone** with lightweight MLP adapters for efficient adapter-only training and inference.
- Built efficient data loaders and a **class-aware sampler** to handle long-tailed distributions (CIFAR100LT / ImageNetLT) and ensure balanced mini-batches during training.
- Added a **per-task auxiliary classifier** (trained with CLIP loss and re-initialized each task) to improve adapter discriminability while keeping inference classifier-free.
- Benchmarked DES against strong baselines, achieving up to **+8.08%** improvement on CIFAR100LT and **+2.56%** on ImageNet.

Exploring Compositionality in Vision Transformers | [Link to Preprint](#) July 2025 - Sept 2025
Lab for Vision and Image Analysis, IIT Hyderabad (Advisor: Dr. Sumohana S. Channappayya)

- Studied how Vision Transformers combine visual components to form image representations, with the goal of improving interpretability and explainability.
- Designed a framework using wavelet transforms to break images into **primitives** (frequency sub-bands) and tested whether their **linear combination** in representation space can approximate the ViT representations.
- Through extensive experiments on ImageNet, showed that despite strong non-linearity, ViTs inherently exhibit compositionality.
- Tested robustness of this framework under **real-world distortions** such as additive noise and image compression.

RISC-V Assembly Code Simulator (C++) | [Link to Code](#) Aug 2024 - Nov 2024
Course Project - Computer Architecture (Dr. Rajesh Kedia)

- Built an **assembly code simulator** to execute RISC-V programs and emulate **registers, memory, stack, and cache** behavior.
- Developed an **assembler** to convert instructions into **machine code** for simulation.
- Implemented **memory segments**: text segment for machine code and data segment for global variables.
- Designed a **stack system** to track function execution and manage local variables.
- Built a **cache simulation** supporting configurable **associativity, block size, and capacity**.

SCHOLARLY ACTIVITY

- Preprint: [Exploring Compositionality in Vision Transformers using Wavelet Representations](#)
- Manuscript under preparation: [Towards Removing Assumptions in Class Incremental Learning](#)
- Poster Presentation at Research Symposium at IITH: [AFCIL work](#)
- Sub Reviewer, AAAI 2026

SCHOLASTIC ACHIEVEMENTS

- IUSSTF-Viterbi Scholar (2025)**: Selected as one of the top 15 research students from across India to conduct research at USC.
- Academic Excellence Award (2024–2025)**: Awarded for achieving the **highest CGPA** in the academic year.
- Dr. Homi Bhabha Young Scientist Competition (Silver Medalist)**: Ranked in the top **0.45%** of participants in the state.
- JEE Mains (2022)**: Scored **99.90 percentile** among 1 million+ candidates nationwide.

Please visit my [portfolio website](#) for more details about my work.